

Historica

(AR/VR Powered Immersive Historical Learning Experience)

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Abstract: The web-based platform of Historica uses Augmented Reality (AR) technology to create new ways for users to learn and experience history. Users can study historical sites and events and ancient civilizations through the platform which combines 3D environments and AI avatars and map navigation features that create interactive learning experiences. The interactive approach of Historica together with its visual elements enables students to comprehend historical material while maintaining better memory retention than conventional history teaching methods which depend on reading static textbooks.

The platform stands out by eliminating the need for specialized hardware such as VR headsets, making it accessible directly through standard web browsers. The system provides access to various devices which different user groups can use, including students, educators, researchers, and history enthusiasts. Users can navigate historical sites while exploring 3D models and accessing supporting materials through user-friendly design. The Time Travel Mode in Historica enables users to view historical sites across various historical periods, while AI-based scene reconstruction technology recreates missing or damaged monuments with precise detail. The platform enables users to engage with Quantum History simulations which let them study alternate historical events and "what-if" scenarios that promote critical thinking and active learning. Historica establishes an educational technology connection by creating a learning platform which students and teachers can use for immersive interactive educational experiences. The program transforms history into an exciting technological experience which improves learning results and protects cultural heritage while it stimulates student interest in history.

Keywords: Augmented Reality (AR), Virtual Reality (VR), Web-Based Platform, 3D Visualization, Artificial Intelligence (AI), Historical Simulation, Digital Heritage Preservation, Interactive Learning, Time Travel Mode, Quantum History.